Software Project Management
Assignment-9
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

When is code review performed during software life cycle?

a. After unit testing
b. After coding and compiling
c. During integration testing
d. During system testing
e. During acceptance testing

Correct Answer: b. After coding and compiling

Detailed Solution:

Code review is a static testing technique performed after coding and compiling but before dynamic testing (such as unit testing). It is done to find defects early by examining the source code without executing it. Performing it later—during or after integration, system, or acceptance testing—would be too late in the lifecycle. Thus, (b) is the correct answer.

QUESTION 2:

Cleanroom technique does not use which one of the following activities?

a. Code walkthrough
b. Code inspection
c. Formal verification for bug removal
d. Testing of code by executing it
e. Statistical quality control

Correct Answer: d. Testing of code by executing it

Detailed Solution:

The Cleanroom software engineering technique emphasizes defect prevention through formal verification (mathematical correctness), code walkthroughs and inspections (static reviews), and statistical quality control (usage-based reliability certification). It deliberately avoids execution-based testing during development—the team does not run the code to find bugs; instead, they rely on formal methods and statistical testing by a separate team for certification. Thus, (d) is the activity not used in Cleanroom.

QUESTION 3:

Which one of the following is true of a code review meeting?

a. It seeks to identify errors, the person(s) responsible for the identified errors, and the solutions for the identified errors
b. It seeks to identify errors and the person(s) responsible for the identified errors, but does not seek solutions
c. It seeks to identify errors, but does not seek solutions or identify the person(s) responsible for the identified errors
d. It seeks to identify errors and their cause as well as outlines the solutions
e. It seeks to identify errors as well as recommends solutions for the same, but does not attribute blame for the error to anyone in particular.

Correct Answer: c. It seeks to identify errors, but does not seek solutions or identify the person(s) responsible for the identified errors

Detailed Solution:

A code review meeting focuses on finding defects in the work product, not on fixing them or assigning blame. The goal is to improve quality by identifying errors, while solutions are developed later during rework, and responsibility is not attributed to individuals—the review is about the product, not the producer. Thus, (c) is the correct statement.

QUESTION 4:

Which of the following charts records and displays the way in which targets have changed throughout the duration of the project?

a. Gantt chart
b. Pie chart
c. Timeline chart
d. PERT chart
e. Flow chart

Correct Answer: c. Timeline chart

Detailed Solution:

A timeline chart records and displays how project targets, such as milestone dates or deadlines, have changed over the course of the project. It visually tracks the evolution of planned versus actual dates, making it useful for monitoring schedule slips. Gantt charts, PERT charts, pie charts, and flow charts do not serve this specific purpose. Thus, (c) is the correct answer according to your book's terminology.

QUESTION 5:

The total value credited to a project at any point is known as

a. the planned value
b. the earned value
c. the actual cost
d. the average cost
e. the optimal cost
f. total value

Correct Answer: b. the earned value

Detailed Solution:

In Earned Value Management (EVM), Earned Value (EV) is the value of work actually completed at a given point in time, representing the budgeted cost of that completed work. Planned Value (PV) is the value of work scheduled, Actual Cost (AC) is what has been spent, and the other options are not standard EVM terms. Thus, (b) is correct.

QUESTION 6:

As part of the implementation of an Employee Payroll Information System project, a number of employee data records have to be manually typed into a database. Which one of the following techniques will be most appropriate for assigning earned value to this project?

a. The 0/100 technique
b. The 50/50 technique
c. The 75/25 technique
d. The 25/75 technique
e. The 100/100 technique
f. The milestone technique
g. The percentage complete technique

Correct Answer: g. The percentage complete technique

Detailed Solution:

Manually typing a large number of employee records into a database is a repetitive, quantifiable task where progress can be measured incrementally (e.g., records completed / total records). The percentage complete technique assigns earned value proportionally as work progresses—if 40% of records are typed, 40% of the budgeted value is earned. Fixed-ratio techniques (0/100, 50/50, etc.) or milestone techniques are better suited for discrete, non-divisible deliverables, not for continuous, measurable tasks like data entry. Thus, (g) is the most appropriate choice.

QUESTION 7:

Suppose the work with a PV of Rs. 55,000 should have been completed by now. Some of that work has not been done, so the EV is only Rs. 45,000. What is the value of Schedule Variance (SV)?

a. Rs. 10,000
b. - Rs. 10,000
c. Rs. 1,00,000
d. Rs. 5,000
e. - Rs. 5,000

Correct Answer: b. - Rs. 10,000

Detailed Solution:

Schedule Variance (SV) is calculated as: $SV=EV-PV$
Given:
EV = Rs. 45,000
PV = Rs. 55,000

$SV=45,000-55,000=-10,000$

QUESTION 8:

Suppose, the work with a PV of Rs. 50,000 should have been completed by now. In fact, some of that work has not been done, so that the EV is only Rs. 40,000. Also, it is given that Rs. 60,000 had actually been spent to get this EV. What is the value of Schedule Performance Index (SPI)?

a. 0.67
b. 0.75
c. 0.80
d. 1.25
e. 1.20
f. 0.83

Correct Answer: c. 0.80

Detailed Solution:

The Schedule Performance Index (SPI) is calculated as: SPI=EV/PV
Given:
EV = Rs. 40,000
PV = Rs. 50,000

SPI=40,000/50,000=0.80

QUESTION 9:

A negative CV means that,

a. the project is over cost
b. the project is below cost
c. the project is as per the planned cost
d. The project is having moderate cost
e. no effect on cost

Correct Answer: a. the project is over cost

Detailed Solution:

In Earned Value Management (EVM), Cost Variance (CV) is calculated as CV = EV – AC. A negative CV means that the actual cost (AC) is higher than the earned value (EV), indicating that the project is over budget (over cost).

QUESTION 10:

Assume that a medium-sized project is experiencing considerable schedule slippage. Which of the followings may not help to get the project back on schedule?

a. Shortening the critical path
b. Altering the activity precedence requirements
c. Speeding up non-critical path activities
d. Reducing scope of the project
e. Reducing the quality of the product

Correct Answer: c. Speeding up non-critical path activities

Detailed Solution:

Speeding up non-critical path activities does not help recover schedule slippage because these activities have slack/float and do not affect the project's overall completion date—only the critical path determines the project duration. Shortening the critical path (a), altering precedence (b), reducing scope (d), or even temporarily reducing quality (e) could potentially shorten the schedule, though reducing quality is unethical and not recommended. Thus, (c) is the action that may not help.

******THE END******